Summarized Paper Reports - Team 3

Paper Name	Publishing Date	Datasets Used	Models & Methods Applied	Results	Limitations & Future Work
Tourism Recommendation System Based on Semantic Clustering and Sentiment Analysis	2020	• TripAdvisor Review Dataset (Egypt + worldwide attractions) • Contains **tourist text reviews + metadata** • Preprocessing: stop-word removal, stemming, tokenization	• **Semantic Clustering**: k-means clustering on TF-IDF and Word2Vec embeddings to group reviews into themes (e.g., history, food, adventure) • **Sentiment Analysis**: polarity classification (positive, negative, neutral) for each cluster • **Matching Process**: user preference profile (clusters + sentiments) compared with attraction profiles to generate ranked recommendations	• **Precision**: ~**81%** • **Recall**: ~**76** % • **F1-score**: ~**78** % • Outperformed baseline collaborative filtering models significantly	• Dataset limited to TripAdvisor reviews (not global scale) • Language dependency: trained only on English reviews • **Cold-start problem** for new users with no review history • No context-awareness (time, season, group size not considered) **Future Work:** Expand dataset to multilingual reviews (Arabic + English), integrate temporal/contextual factors, include multimodal data (photos, check-ins, ratings), build a real-time mobile app prototype
Optimisation Using Natural Language Processing: Personalized Tour Recommendation for Museums	Jan 2015	Musée de l'Orangerie (Paris) dataset • **144 artworks** across **10 rooms** • Data fields: artist, title, year, textual description	• **NLP (Enertex)**: Textual energy method to score "intrinsic importance" of artworks from descriptions • **Visitor Preference Modeling**: user interests represented as vectors (e.g., preferred artists), similarity measured with cosine similarity • **Optimization (ILP – Integer Linear Programming)**: models museum layout as graph, finds optimal tour path under time limits • **Scoring Functions Tested**: f1 = equal scores, f2 = Enertex-based, f3 = visitor preference-based, f4 = hybrid (weighted Enertex + preference)	• Evaluation metric: **Relevance Percentage (rp)** (how well tours match user preferences) • **Hybrid model (f4)** improved tour relevance by up to **40%** compared to baseline (f1) • Optimal tours designed for **~150 minutes (2.5 hours)** showed the highest visitor satisfaction • f2 (Enertex) worked best for discovery-oriented visitors, f3 (preference- based) best for strongly biased tastes, and f4 (hybrid) best overall	• Tested only on a **single museum dataset** (limited scalability) • Visitor preferences restricted to **artists only**, not broader themes/styles • No contextual factors (e.g., group dynamics, accessibility, real-time changes) **Future Work:** Extend personalization to themes/styles, integrate advanced NLP (topic modeling, sentiment analysis), develop **mobile/voice-guided tours**, and scale optimization for larger museums with thousands of artworks

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<td>A Deep Learning Approach to Ancient Egyptian Hieroglyphs Classification</td>
<td>2021</td>
<td>•A public dataset extracted from the Pyramid of Unas, containing 4310 labeled grayscale images of hieroglyphs, annotated according to Gardiner’s sign list. This dataset offers a standardized benchmark with relatively clean images.
• A custom dataset compiled by the authors from multiple supports (stone, papyrus, etc.), reflecting more realistic conditions with noise and erosion. This dataset was chosen to test the models’ robustness in challenging, real-world scenarios.
However, it is important to note that neither of these datasets is publicly available. The first one, extracted from the Pyramid of Unas, has not been released beyond the scope of the paper, while the second was privately compiled by the authors from various supports and remains inaccessible.
As a practical alternative, researchers may refer to publicly accessible datasets such as the Egyptian Hieroglyphs dataset on Hugging Face , which provides a comparable collection of labeled hieroglyph images. Although not identical, this dataset can serve as a useful proxy for experimentation and benchmarking.</td>
<td>• ResNet-50: Developed by Microsoft Research with 23M+ parameters. • Uses residual blocks to solve vanishing gradient issues. • Processes 224x224 images.

• Inception-v3: Developed by Google with 42 layers Uses inception modules to capture features at multiple scales. • Processes 299x299 images

• Xception: Evolution of Inception model using depthwise separable • More efficient with 299x299 images.

• Glyphnet (Proposed Network): Inspired by Xception, but tailored for hieroglyphs Uses separable convolutions, batch normalization, dropout, and global average pooling. Only ~0.5M parameters compared to 20M+ in others Faster, lighter, less prone to overfitting.</td>
<td>TABLE 1. Networks performance using the transfer learning approach
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<tr><th>Architecture</th><th>Accuracy</th><th>Precision</th><th>Recall</th><th>F1-Score</th></tr>
<tr><td>ResNet-50</td><td>0.906</td><td>0.882</td><td>0.825</td><td>0.840</td></tr>
<tr><td>Inception-v3</td><td>0.864</td><td>0.730</td><td>0.737</td><td>0.717</td></tr>
<tr><td>Xception</td><td>0.834</td><td>0.715</td><td>0.720</td><td>0.703</td></tr>
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TABLE 2. Networks performance using training from scratch
<table>
<tr><th>Architecture</th><th>Accuracy</th><th>Precision</th><th>Recall</th><th>F1-Score</th></tr>
<tr><td>ResNet-50</td><td>0.945</td><td>0.919</td><td>0.905</td><td>0.903</td></tr>
<tr><td>Inception-v3</td><td>0.948</td><td>0.917</td><td>0.904</td><td>0.900</td></tr>
<tr><td>Xception</td><td>0.956</td><td>0.919</td><td>0.930</td><td>0.919</td></tr>
<tr><td>Glyphnet</td><td>0.976</td><td>0.975</td><td>0.965</td><td>0.968</td></tr>
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Contributions of the Work:
• Applied deep learning methods to hieroglyph classification, a novel area in Egyptology.
• Compared multiple CNN architectures on two datasets of hieroglyph images.
• Proposed Glyphnet , a new CNN optimized for this task.
• Demonstrated superior performance of Glyphnet in bothaccuracy and speed.
• Laid foundations for future AI-based tools in automatic recognition, classification, and translation of hieroglyphs</td>
<td>• Limited datasets: Only 4309 images across 40 categories, relatively small compared to large-scale benchmarks.
• Class imbalance: Some hieroglyph categories had far fewer images.
• Computational resources: Although Glyphnet is lighter, CNN training still requires GPUs.
• Focus on single-character classification: Full text recognition and translation remain future challenges.

Future Directions:
• Extending classification from single hieroglyphs to full words and connected text.
• Integrating the classifier into a full OCR pipeline that can support transliteration and translation.
• Enhancing robustness against damaged or eroded inscriptions.
• Leveraging larger linguistic and image datasets for broader Egyptological applications.</td>
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A BERT-Based Intelligent Chatbot for Tourism Information and Recommendation in Vietnam	2023	• **TWCS (Customer Support on Twitter) Dataset**: A large corpus of customer service tweets. [Available on Kaggle] • **Tourism Subset**: A processed, hand-labeled subset of 7,319 dialogue samples extracted from airline Twitter feeds (American Air, British Airways, Southwest Air) relevant to travel inquiries. • **Labels**: 11 user intents (e.g., greet, goodbye, get_suggest, give_city, give_interests). • **Preprocessing**: Data was split for 5-fold cross-validation (80% training, 20% testing).	• **Intent Classification**: A comparative analysis of three NLP models: Bag-of-Words, BERT [Original Paper], and RoBERTa [Original Paper]. • **Chatbot Framework**: Built using the **RASA** open-source framework [RASA Official Site] for dialogue management. • **System Architecture**: A comprehensive architecture featuring a Flutter/Dart mobile app, a React-based content management web app, a **PostgreSQL** database [PostgreSQL Official Site], and RASA NLU/action servers. • **Deployment**: The trained BERT model was integrated into a production-ready Android application.	• **BERT** achieved the highest performance after 200 training epochs: - **Accuracy**: **90.0%** - **F1-Score**: **90.1%** • The deployed chatbot application successfully handled predefined scenarios for greetings, information retrieval, and place recommendations with high accuracy. • The system significantly outperformed the Bag-of- Words baseline and the RoBERTa model on this specific intent classification task.	• **Dataset Limitations**: Trained on a domain-specific subset of customer service data, which may not encompass the full range of tourist queries or conversational nuances. • **Language Dependency**: The system is designed only for English, limiting its utility for the large number of non- English speaking tourists in Vietnam. • **Context-Awareness**: Does not incorporate real-time contextual factors like weather, time, or dynamic local events. **Future Work**: Expand the training dataset with more diverse tourist intents and Vietnamese language support, integrate real-time contextual and multimodal data (e.g., images, real-time reviews), and enhance the personalization of the recommendation engine.
Image Based Hieroglyphic Character Recognition	November 2018.	• A dataset of about **700 query images** was used. • It covered **197 different hieroglyphic characters**, organized according to Gardiner's codes. • The dataset also included different orientations (left/right) for some symbols to support correct recognition.	• OCR-based pipeline using **Image Processing**. • **Image Acquisition**: grayscale conversion, Canny edge detection, Hough Transform. • **Segmentation**: bounding box + connected components labeling, preserving reading order (top-to-bottom, right-to-left when applicable). • **Post-processing**: noise removal, resizing, filtering out small images. • **Feature Extraction & Matching**: Histogram of Oriented Gradients (HOG) + Euclidean distance. • **English Mapping**: Gardiner's codes mapped to English translations. • Alternative techniques tested: Template Matching, SURF (lower accuracy).	• **Segmentation accuracy**: avg. 84.33% compared to manual segmentation. • **Recognition accuracy**: avg. 66.64%. • Outperformed state-of-the- art Chinese character recognition methods applied to hieroglyphs (55.27%). • Produced meaningful English translations, though not grammatically structured.	• **Limitations**: – Translated English lacks proper grammar/syntax. – Mainly supports vertical column inscriptions; not generalized to all hieroglyphic layouts. – Low-quality/blurry hieroglyphs reduce accuracy. • **Future Work**: – Integrate **NLP** for grammatical English translation. – Develop a **mobile application** for tourists/researchers. – Expand dataset for broader coverage of hieroglyphic formats.

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Recognition of Egyptian hieroglyphic texts through focused generic segmentation and cross-validation voting	2025	**Dataset** **New dataset created by authors:** • **Size:** 13,729 hieroglyph images labeled. • **Classes:** 310 classes (later filtered to 164 with ≥7 samples/class for experiments). • **Covers:** Old, Middle, and New Kingdoms (≈3000 years). • **Format:** Cropped and standardized images from scanned stelae • **Availability:** Dataset and models are made available to the scientific community	**Segmentation:** • **MBRS (Method Based on Region Segmentation) → classic computer vision (Canny edges).** • **IGSM (Integrated Generic Segmentation Model) → based on SAM (Segment Anything Model)** **Classification:** • **ConvNeXt backbone CNNs.** • **Cross-Validation Voting (CVV): ensemble method with both Soft Voting (CVV-SV) and Hard Voting (CVV-HV)** **Training details:** • **90% train/validation, 10% test split.** • **Data augmentation: shifts, rotations, scaling, flips.** • **Final training set expanded to 18,718 images**	**Their method (IGSM + CVV):** • **Accuracy: up to 0.95** • **F1-score: 0.91** • **Balanced Accuracy: 0.94–0.99** **Comparison with other approaches (Table 8):** • **HOG+SDM (Franken): 0.855 accuracy** • **Detectron2+Glyphnet (Barucci): 0.934 accuracy** • **R-CNN+SiameseNN (Sobhy): 0.885 accuracy** • **Their method (IGSM+CVV-SV): 0.940 accuracy, 0.887 F1-score → best performance** **Conclusion: Their method clearly outperforms the state-of-the-art detection-based approaches in hieroglyph recognition.**	**Dataset expansion:** • **Add more hieroglyphs for broader coverage.** • **Increase diversity of classes to better capture complexity.** **More media types: Extend recognition beyond stelae to manuscripts, papyri, and other artifacts** **Translation step: Move beyond recognition to automatic transliteration and translation into modern languages (e.g., English).** **Model robustness: Improve handling of noise, erosion, and diverse writing style**
Deep Learning Meets Egyptology: A Hieroglyphic Transformer for Translating Ancient Egyptian	2024	**Datasets Used** • **Thesaurus Linguae Aegyptiae (TLA)** — main dataset of ancient Egyptian texts https://thesaurus-linguae-aegyptiae.de • **TLA Snapshot (2018, CC BY-SA 4.0 International)** — the exact dataset version used in the paper https://nbnresolving.org/urn:nbn:de:kobv:b4-opus4-29190 **Dataset details:** – 103,906 total data points → filtered to **61,605** (Old & Middle Egyptian) – Includes: Gardiner codes, transliterations, German/English translations, lemma IDs, POS tags, inflection codes, and metadata	• **Source Code:** **GitHub Repository — Hieroglyphic Transformer** **(Contains training scripts, data cleaning methods, evaluation metrics, and pre-trained model configuration.)** • **Model: Fine-tuned M2M-100 Transformer (multilingual model by Facebook AI)** • **Approach: Transfer learning applied to a low-resource ancient language (Egyptian)** • **Data Cleaning: Over 280 custom preprocessing rules for text normalization, symbol correction, and removal of inconsistencies** • **Training Setup:** **– 11 models trained (6–20 epochs each)** **– Optimizer: Adam** **– Learning rate: 3e-5** **– Batch size: 16 (180 for the final "ALL" model)** **– Early stopping: triggered after 15–20 non-improving evaluations** • **Evaluation Metrics:** **– Automatic: SacreBLEU, RougeL** **– Manual: Human evaluation (grammar,**	• After cleaning: BLEU ↑ from 4.0 → 54.4, RougeL ↑ from 18.4 → 62.8 • Best model (ALL): SacreBLEU up to 61.5 (DE→EN), RougeL ~67.7 • Accurate translations from hieroglyphs → German/English • Human tests: performs well on short/medium sentences (1–2 clauses) • Strong lemma and POS tag prediction	• Low-resource limitation: limited English data; mainly German translations • Struggles with long texts (3+ clauses) or rare terms • No contextual/temporal factors (time, domain, genre not modeled) • Future work: Expand to Late Egyptian, Demotic, Coptic; add more languages; integrate multimodal data (images + text); improve robustness for long passages

			morphology, semantics, coherence) **• Data Augmentation:** **– Back-translation from German → English using M2M-100 to expand English data**		